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EFFICIENT DEEP LEARNING

Optimizing image classification performance with ResNet18

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Introduction

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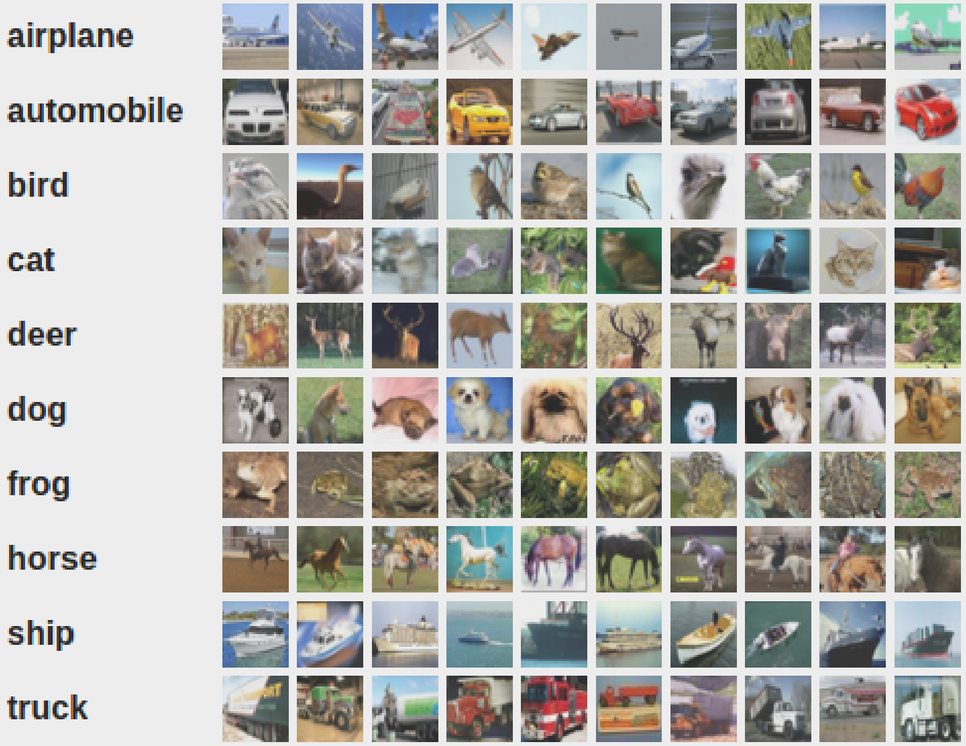
Reach 90% accuracy while minimizing the score of the model.

CIFAR10



60.000 color images
Size: 32 × 32 pixels
10 classes, with 6.000 images per class

- 50.000 for training
- 10.000 for testing

Baseline model (ResNet18)

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Baseline model

ResNet18

Epochs: 200
Batch_size: 32
Optimizer: SGD
Learning rate init: 0.1
Scheduler: CosineAnnealingLR
Loss function: CrossEntropyLoss

Test acc: 94.94%

Methodology and experiments

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Score

$$\text{score} = \underbrace{\frac{[1-(p_s+p_u)]\frac{q_w}{32}w}{5.6\cdot 10^6}}_{param} + \underbrace{\frac{(1-p_s)\frac{\max(q_w,q_a)}{32}f}{2.8\cdot 10^8}}_{ops}$$

Where:

- p_s: structured pruning
- p_u: unstructured pruning
- q_w: quantization of weights
- q_a: quantization of activations
- w: number of weights
- f: number of mult-adds (MACs) operations
- $5.6\cdot 10^6$ and $2.8\cdot 10^8$ are the reference param and ops scores of the ResNet18 network in half precision.

Accuracy

If the score is promising, launch training to observe the accuracy of a certain configuration.

```
>> from thop import profile
# Profile the model
macs, params = profile(model, inputs=(dummy_input,))
```

Methodology

Reduce the number of parameters

- **Width scale** (channel multiplier = **0.75**)
  - Change to another version of ResNet with less parameters, such as ResNet20
- Pruning (Experiments on lr=0.01)
  - **Structed**
  - **Unstructed**
- Quantization
  - Binary Connect
  - XNOR-Net
  - **FP16**
  - **FP32**
- Factorization
  - Spatial
  - **Depthwise**
  - Grouped

Quantization

Width scale

Factorization

$$\text{score} = \underbrace{\frac{[1-(p_s+p_u)]\frac{q_w}{32}w}{5.6\cdot 10^6}}_{param} + \underbrace{\frac{(1-p_s)\frac{\max(q_w,q_a)}{32}f}{2.8\cdot 10^8}}_{ops}$$

Pruning

Improve accuracy

- Data augmentation
  - AutoAugment
  - RandomErase
  - RandomHorizontalFlip
  - RandomCrop
  - **CutMix** (alpha = **1.0**)
- Knowledge Distillation
  - **NTCE-KD**

Alternative approaches evaluated

Model choice

	DenseNet	ResNet18	ResNet20
Duration of training	~22 hours	~4 hours	~6 hours
Accuracy	93.79%	94.94%	91.96%

Choice of ResNet18 due to time constraints and its better performance results.

Data split

- Train
- Validation

VS

- Train (80% of training datase)
- Validation (20% of training datase)
- Test (100% of test dataset)

Alternative approaches evaluated

Quantization

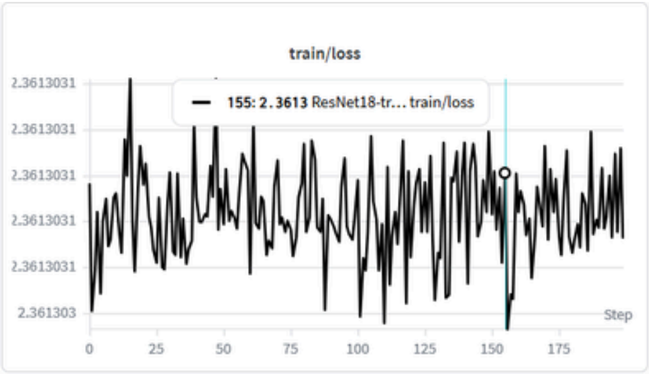
	Binary Connect	XNOR-Net
Weights	1 bit*	1 bit
Activations	32 bits	1 bit

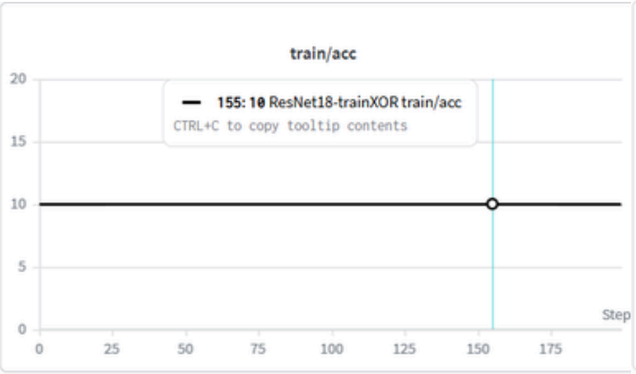
* First and last layers are not binarized not to compromise the performance.

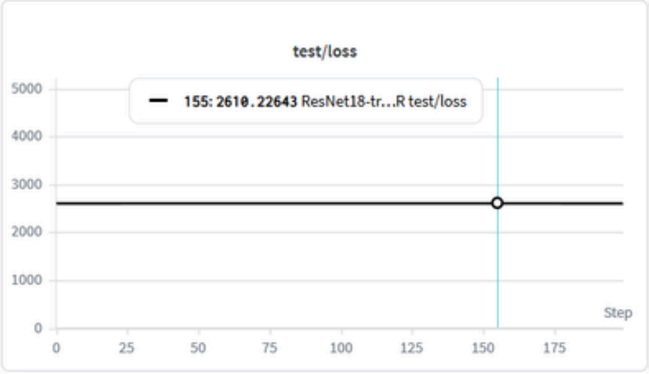
Experiments on lr=0.01

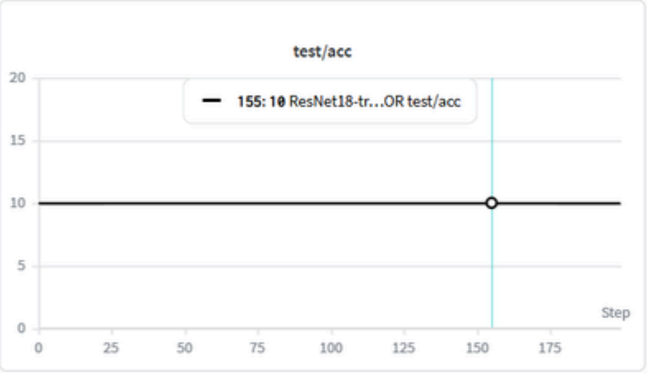
Ps=0, Pu=0, qw=1, qa=1
Score=0.124631
Accuracy = ?

Unstable train, with accurancy always around 10% = random guess

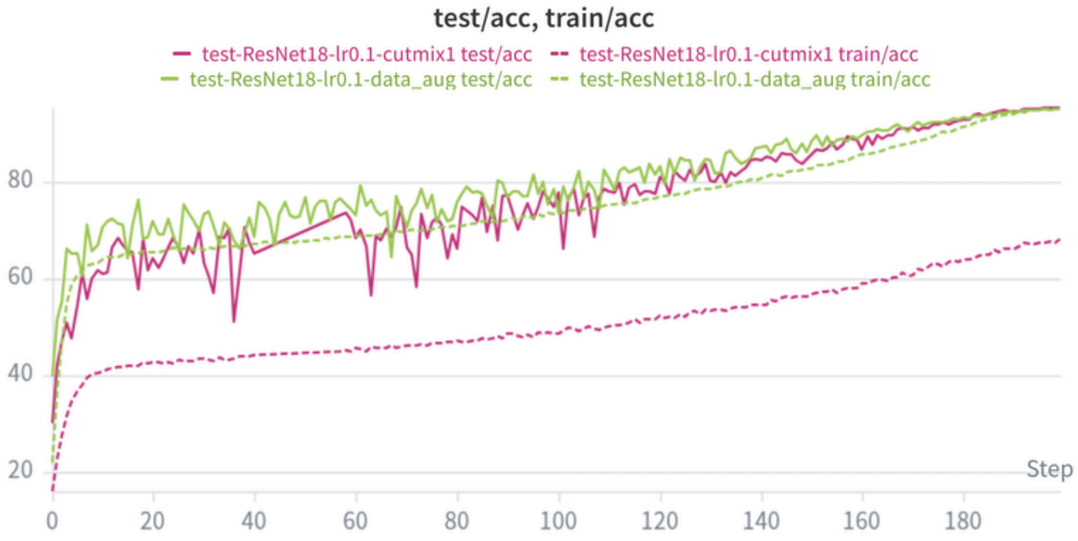








Cutmix



95.38 test-ResNet18-lr0.1-cutmix1 test/acc + 5.58

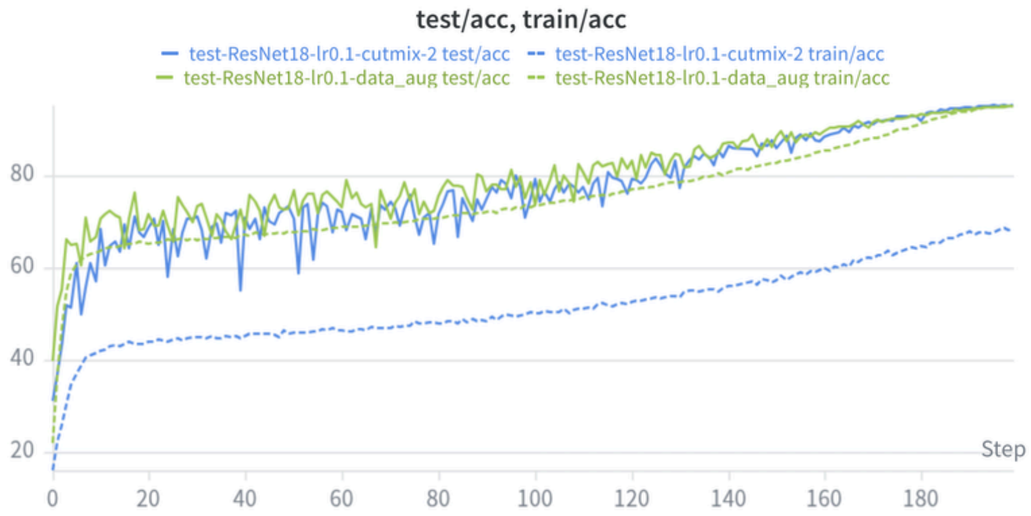
95.08 test-ResNet18-lr0.1-data_aug test/acc + 5.28

95.074 test-ResNet18-lr0.1-data_aug train/acc

68.30413 test-ResNet18-lr0.1-cutmix1 train/acc

RandomErasing p=0.25, scale=0.02, 0.2, ratio=0.3, 3.3
AutoAugment
RandomHorizontalFlip
RandomCrop

Cutmix



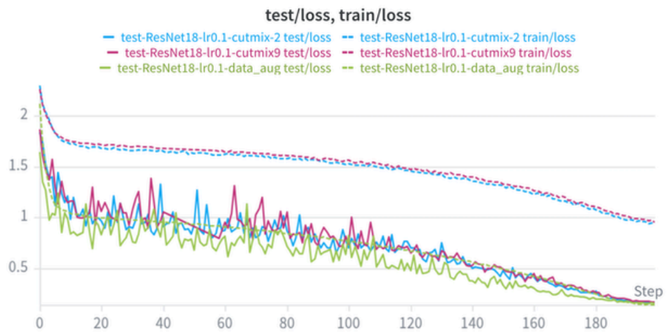
95.42 test-ResNet18-lr0.1-cutmix-2 test/acc + 5.62

95.08 test-ResNet18-lr0.1-data_aug test/acc

95.074 test-ResNet18-lr0.1-data_aug train/acc

68.51976 test-ResNet18-lr0.1-cutmix-2 train/acc

AutoAugment
RandomHorizontalFlip
RandomCrop
Cutmix



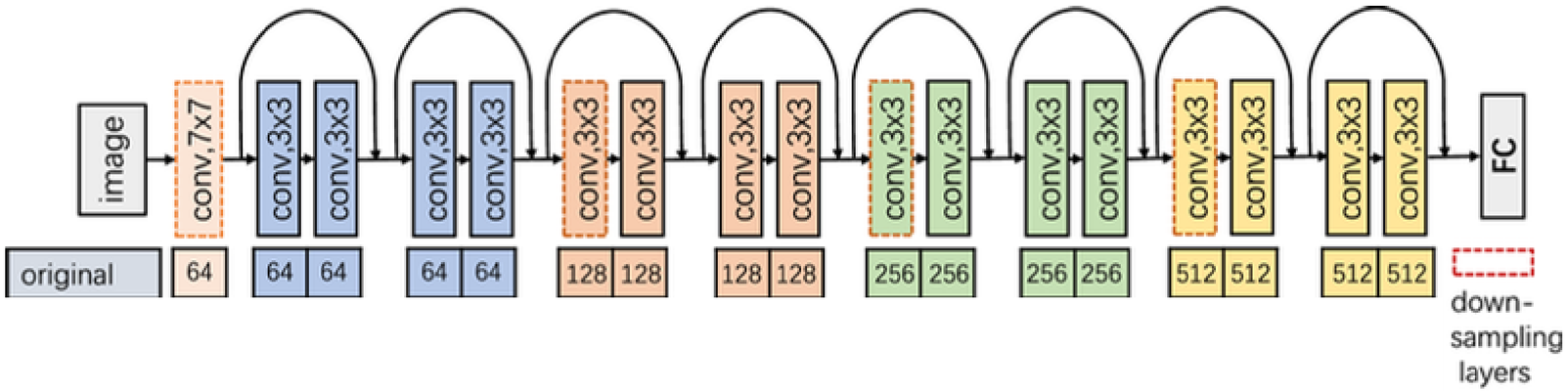
Best approaches applied

Width scale

Channel multiplier: 0.75
Accuracy: 94.03%

Uniformly scale the number of channels (depth) across all layers.

More channels → the network learns more complex patterns (filters)
Fewer channels → reduce the model's footprint

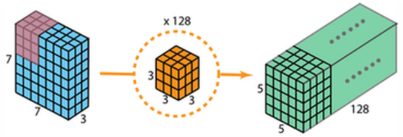


Best approaches applied

Factorization

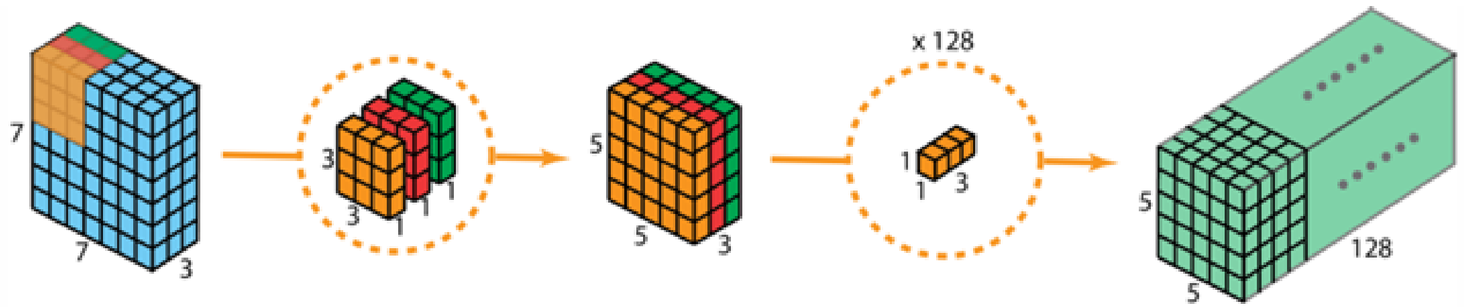
-	Accuracy	Score
Depthwise	91.94%	0.526743
Spatial	91.24%	3.044062
Grouped	93.38%	1.050378

Without Depthwise



$$N^N_{mul} = H_{out} \cdot W_{out} \cdot h \cdot w \cdot D_{in} \cdot D_{out} = 5 \cdot 5 \cdot 3 \cdot 3 \cdot 3 \cdot 128 = 86400$$

Depthwise Separable Convolution

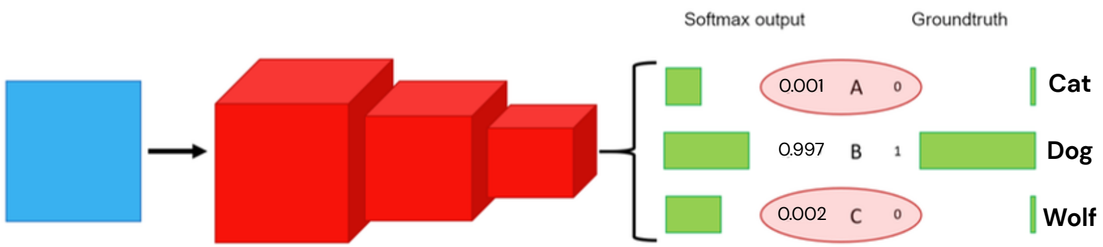


- Depthwise: One filter per input channel (3x3)
- Pointwise: 1x1 convolution to combine channel information

$$N^D_{mul} = H_{out} \cdot W_{out} \cdot h \cdot w \cdot D_{in} \cdot 1 + H_{out} \cdot W_{out} \cdot 1 \cdot 1 \cdot D_{in} \cdot D_{out}$$

$$N^D_{mul} = 5 \cdot 5 \cdot 3 \cdot 3 \cdot 3 \cdot 1 + 5 \cdot 5 \cdot 1 \cdot 1 \cdot 3 \cdot 128 = 10275$$

Non-Target-Class-Enhanced Knowledge Distillation (NTCE-KD)



Existing methods fail to fully exploit the information in non-target classes.

Method	Knowledge	Teacher (baseline)	Student (baseline)	Accuracy	Code
FitNet (Romero et al., 2014)	Logit	ResNet-34 (73.31)	ResNet-18 (69.75)	70.31 (+0.56)*	Link
KD (Hinton, 2015)	Logit	ResNet-34 (73.31)	ResNet-18 (69.75)	70.62 (+0.87)*	Link
		ResNet-50 (76.16)	MobileNetV1 (68.87)	68.58 (-0.29)*	
AT (Zagoruyko & Komodakis, 2016)	Feature	ResNet-34 (73.31)	ResNet-18 (69.75)	70.30 (+0.55)*	Link
		ResNet-50 (76.16)	MobileNetV1 (68.87)	69.56 (+0.69)*	
CRD (Tian et al., 2019)	Feature	ResNet-34 (73.31)	ResNet-18 (69.75)	71.17 (+1.42)	Link
		ResNet-50 (76.16)	MobileNetV1 (68.87)	71.37 (+2.50)	
OFD (Heo et al., 2019a)	Feature	ResNet-34 (73.31)	ResNet-18 (69.75)	70.81 (+1.06)	Link
		ResNet-50 (76.16)	MobileNetV1 (68.87)	71.25 (+2.38)	
SphericalKD (Guo et al., 2020a)	Logit	ResNet-34 (73.31)	ResNet-18 (69.75)	72.30 (+2.55)	Link
RKD (Chen et al., 2021d)	Feature + Logit	ResNet-34 (73.31)	ResNet-18 (69.75)	71.61 (+1.86)	Link
		ResNet-50 (76.16)	MobileNetV1 (68.87)	72.56 (+3.69)	
NTCE-KD (Li et al., 2024a)	Logit	ResNet-34 (73.31)	ResNet-18 (69.75)	73.12 (+3.37)	–
		ResNet-50 (76.16)	MobileNetV1 (70.13)	74.90 (+6.03)	

A Comprehensive Survey on Knowledge Distillation, 2025

Why Use NTCE-KD ?
Improve knowledge distillation by enhancing non-target class information

$$L_{total} = \alpha L_{CE} + \beta L_{KL}$$

$$\mathcal{S}_{n0} = \mathbf{v}_n^{(y_n)} - \max_{k\in[1,K],k\neq y_n} \mathbf{v}_n^{(k)}.$$

the difference between the target class and the maximum non-target class

$$\tilde{\mathbf{v}}_n^{(y_n)} = \mathbf{v}_n^{(y_n)} - \mathcal{S}_n,$$

shrink the target class in the logits and apply softmax to normalize them

$$\tilde{\mathbf{v}}_{n,m}^{(y_n)} = \mathbf{v}_n^{(y_n)} - \lambda_m * \mathcal{S}_{n0}.$$

a shrinkage coefficient

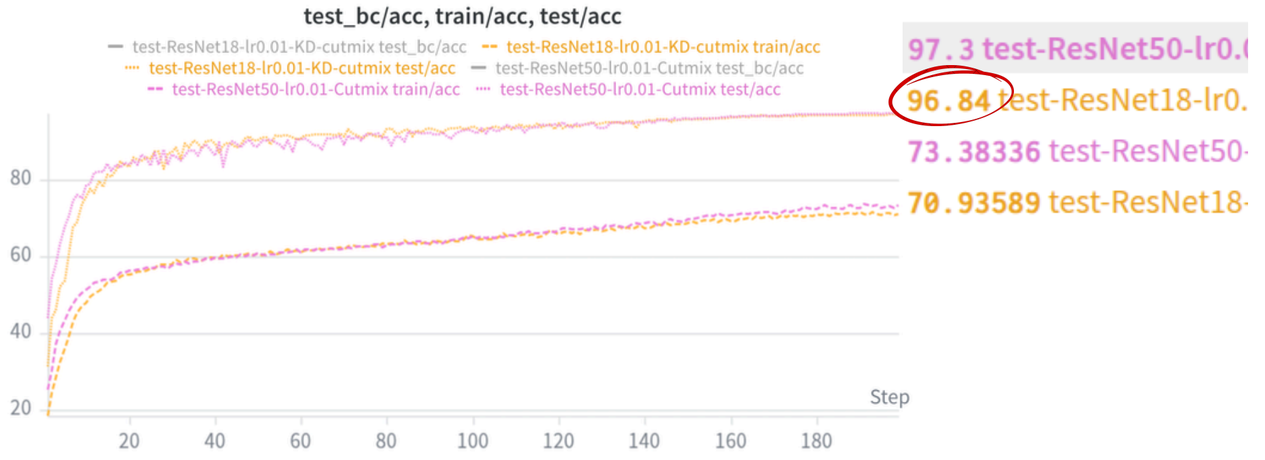
NTCE-KD: Non-Target-Class-Enhanced Knowledge Distillation, 2024

λ	Modified Dog Logit	New Logits [C, D, W]	Softmax Output [C, D, W]
0	9.0 − (0.0 × 6.5) = 9.0	[2.0, 9.0, 2.5]	[0.001, 0.997, 0.002]
0.25	9.0 − (0.25 × 6.5) = 7.375	[2.0, 7.375, 2.5]	[0.004, 0.989, 0.007]
0.5	9.0 − (0.5 × 6.5) = 5.75	[2.0, 5.75, 2.5]	[0.02, 0.93, 0.05]
0.75	9.0 − (0.75 × 6.5) = 4.125	[2.0, 4.125, 2.5]	[0.08, 0.74, 0.18]
1.0	9.0 − (1.0 × 6.5) = 2.5	[2.0, 2.5, 2.5]	[0.27, 0.36, 0.36]

Best approaches applied

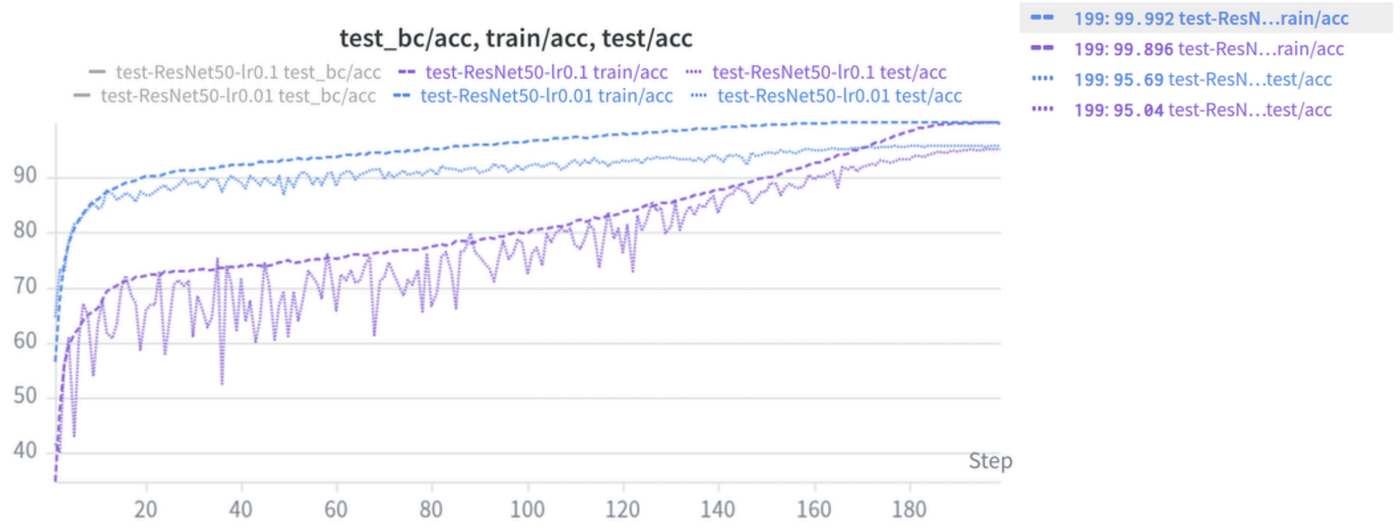
Non-Target-Class-Enhanced Knowledge Distillation (NTCE-KD)

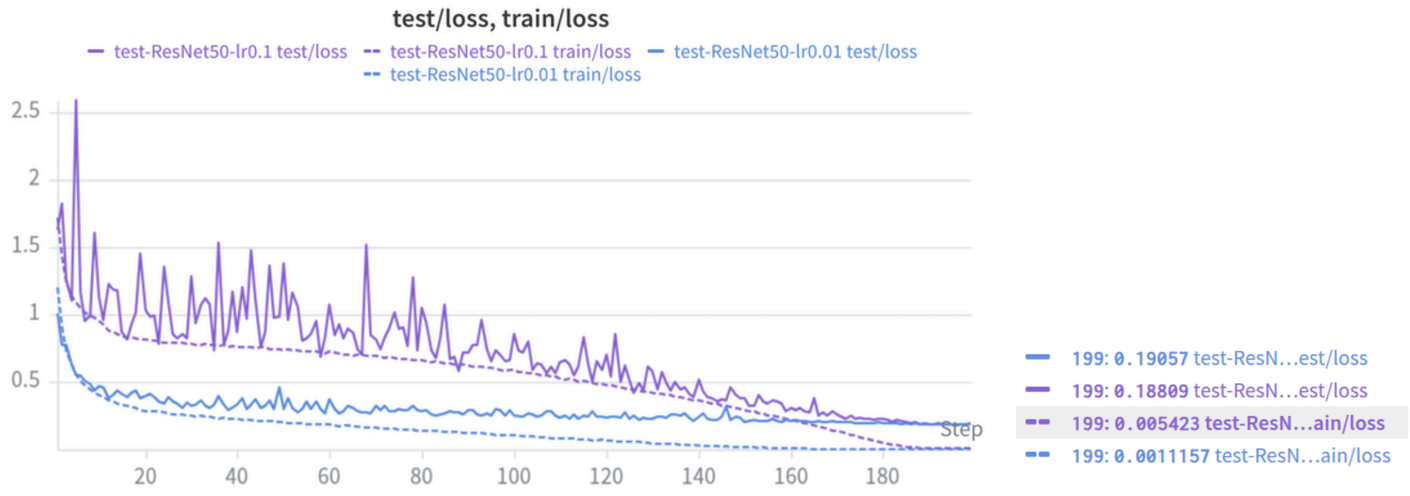
Teacher: ResNet50
Student: ResNet18
Epochs: 200
Batch_size: 32
Optimizer: SGD
Learning rate init: 0.01
Scheduler: CosineAnnealingLR
Standard Data Augmentation
AutoAugment
Cutmix
Loss function: NTCE_KD_Loss
temperature=4.0
alpha=1.0
beta=8.0



Knowledge Distillation

Model: Resnet50
Epochs: 200
Batch_size: 32
Optimizer: SGD
Learning rate init: 0.1
Learning rate init: 0.01
Scheduler: CosineAnnealingLR
Loss function: NTCE_KD_Loss
temperature=4.0
alpha=1.0
beta=8.0





Results and performance analysis

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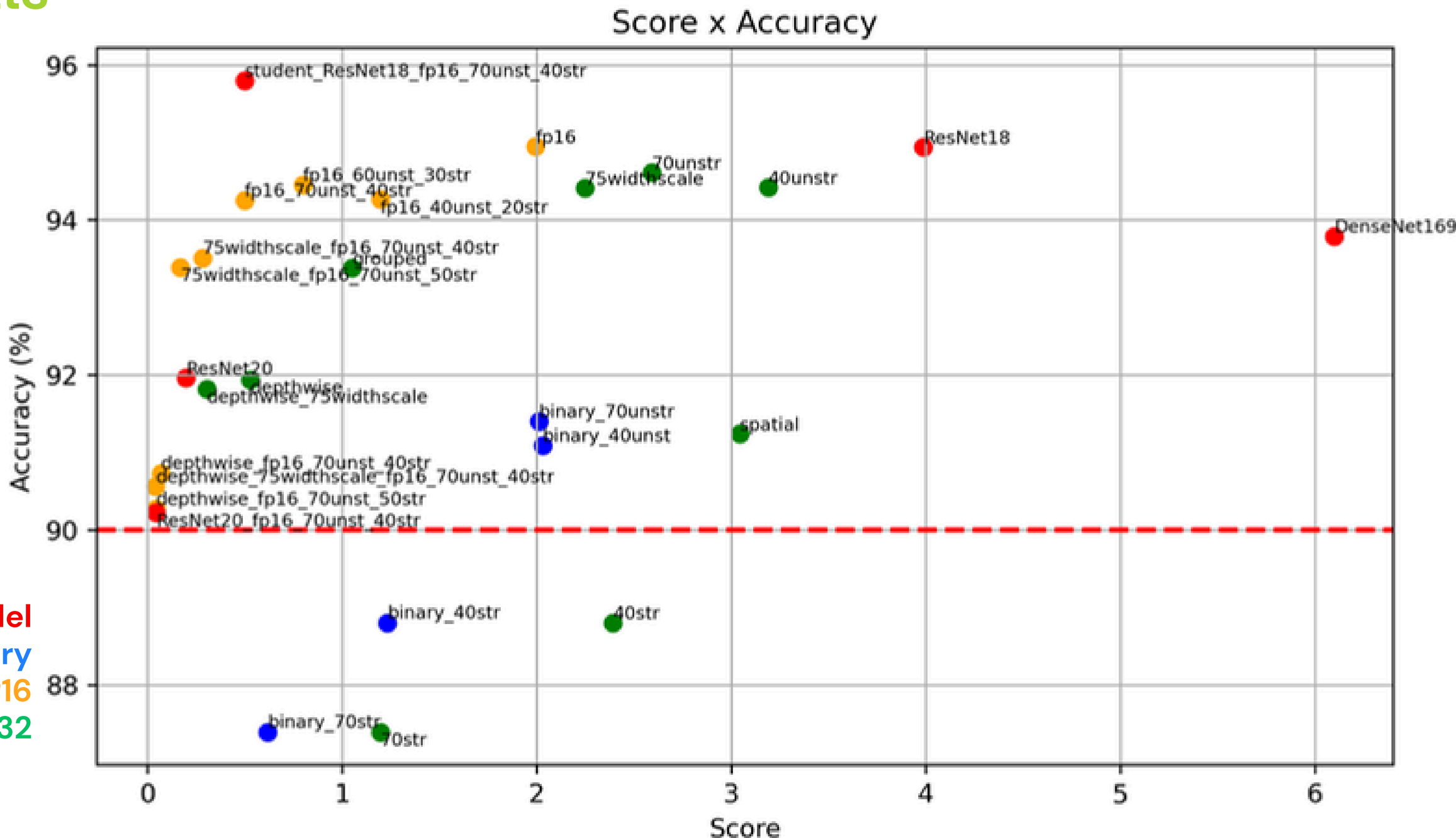
Results



Basic model
Binary
FP16
FP32

Best results

Lowest scores with accuracy uper than 90%

Configuration	Score	Accuracy
ResNet20, FP16, 70% Unstructured + 40% Structured	0.04215	90.22%
ResNet18, Deptwise, FP16, 70% Unstructed + 50% Structed	0.041709	90.27%
ResNet18, Depthwise, 75% Width scale, FP16, 70% Unstructed + 50% Structed	**0.039884**	**90.56%**
Student ResNet18, FP16, Cutmix 70% Unstructured + 40% Structured	**0.498090**	**95.80%**

Other results


```
===== SCORES =====
Basic model (no quant, no prune): 3.988208
No quant, 40% Unstructured: 3.190067
No quant, 70% Unstructured: 2.591462
No quant, 40% Structured: 2.392925
No quant, 70% Structured: 1.196462
Width Scale 75%: 2.246512
Depthwise: 0.526743
Spatial: 3.044062
Grouped: 1.050378
Depthwise, Width Scale 75%: 0.303490
Binary, 40% Unstructured: 2.030270
Binary, 70% Unstructured: 2.011564
Binary, 40% Structured: 1.233127
Binary, 70% Structured: 0.616564
FP16, no pruning: 1.994104
FP16, 40% Unstructured + 20% Structured: 1.196213
FP16, 60% Unstructured + 30% Structured: 0.797268
FP16, 70% Unstructured + 40% Structured: 0.498090
Width Scale 75% + FP16 + 70% Unstructured + 40% Structured: 0.280926
Width Scale 75% + FP16 + 70% Unstructured + 50% Structured: 0.168600
Depthwise, FP16, 70% Unstructured + 40% Structured: 0.068046
Depthwise, FP16, 70% Unstructured + 50% Structured: 0.041709
Depthwise, Width Scale 75%, FP16, 70% Unstructured + 40% Structured: 0.039884
DenseNet169: 6.100940
ResNet20: 0.197285
ResNet20, FP16, 70% Unstructured + 40% Structured: 0.042156
Student ResNet18, FP16, 70% Unstructured + 40% Structured: 0.498090
```

Conclusion

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Conclusion

- The techniques applied allow us to reduce the score of the baseline model keeping the accuracy upper than 90%.
- The structured pruning has a huger impact on the score value than the unstructed one, so for more than 40% the accuracy decreases from 90%. These differences were reduced on FP16.
- Even with more parameters than other ResNet architectures, such as ResNet 20, it was possible to decrease the number of parameters of the ResNet 18 significantly with the widthscale and factorization.
- The knowledge distillation greatly helps improve model accuracy, even under compression constraints. The student ResNet18 achieved a significant accuracy improvement (95.80%) compared to all compressed models.
- The FP16 quantization got the best equilibrium between the accuracy and score.

Opportunities for future work

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Opportunities for future work

Possible scenarios:

- Apply CutMix on ResNet18 with: Depthwise, 75% Width scale, FP16, 70% Unstructed + 50% Structed
- Apply depthwise on Student ResNet18 with: FP16, Cutmix, 70% Unstructured + 40% Structured
- Test FP8 quantization to further reduce memory usage. Compare FP16 vs FP8 in terms of score and accuracy.
- Explore the ResNet 20 with the same sonfigurations as we applied for ResNet 18.

Merci beaucoup pour votre attention !

Avez-vous des questions ?

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